

HAO ZHOU

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🌐 hzhou3.github.io

📍 Mountain View, CA

RESEARCH AREAS

AI-Powered Mobile and Wearable Systems · Human-Centered AI · AI for Health · IoT/CPS

EDUCATION

The Pennsylvania State University

Doctor of Philosophy in Computer Science and Engineering

2021 - 2026

Advisor: Prof. Mahanth Gowda

The University of Mississippi

Master of Science in Computer Science

2019 - 2021

Bachelor of Science in Computer Science

2016 - 2019

PUBLICATIONS

ICML 26

Physiology-Aware Masked Cross-Modal Reconstruction for Biosignal Representation Learning.

Hao Zhou, Simon A. Lee, Cyrus Tanade, Keum San Chun, Juhyeon Lee, Migyeong Gwak, Megha Thukral, Justin Sung, Eugene Hwang, Mehrab Bin Morshed, Li Zhu, Viswam Nathan, Md Mahbubur Rahman, Subramaniam Venkatraman, Sharanya Arcot Desai

MobiCom 26

Exploring the Feasibility of Full-Body Muscle Activation Sensing with Insole Pressure Sensors.

Hao Zhou, Mahanth Gowda

ICASSP 26

A Personalized Real-time Proactive Voice Memory Assistant.

Hao Zhou, Md Mahbubur Rahman, Simon A Lee, Baiying Lu, Juhyeon Lee, Cyrus Tanade, Megha Thukral, Md Sazzad Hissain Khan, Samsad Ul Islam, Subramaniam Venkatraman, Sharanya Arcot Desai

ICLR 26

HiMAE: Hierarchical Masked Autoencoders Discover Resolution Specific Structure in Wearable Time Series.

Simon Lee, Cyrus Tanade, **Hao Zhou**, Juhyeon Lee, Megha Thukral, Minji Han, Rachel Choi, Md Sazzad Hissain Khan, Baiying Lu, Migyeong Gwak, Mehrab Bin Morshed, Viswam Nathan, Md Mahbubur Rahman, Li Zhu, Subramaniam Venkatraman, Sharanya Arcot Desai

NeurIPS 26

TS4H

Towards On-device Foundation Models for Wearable Signals.

Simon Lee, Cyrus Tanade, **Hao Zhou**, Juhyeon Lee, Megha Thukral, Minji Han, Baiying Lu, Sharanya Desai

MobiSys 25 Rising Star	<i>Rethinking Inexpensive Wearables in the Era of AI: From Motion Analytics to Mobile Health.</i> Hao Zhou
ICASSP 25	<i>Know Your Heart Better: Multimodal Cardiac Output Monitoring Using Earbuds.</i> Hao Zhou , Md Mahbubur Rahman, Mehrab Bin Morshed, Yunzhi Li, Md Saiful Islam, Larry Zhang, Jungmok Bae, Christina Rosa, Wendy Berry Mendes, Jilong Kuang
UbiComp 25	<i>WindDancer: Understanding Acoustic Sensing under Ambient Airflow.</i> Kuang Yuan, Dong Li, Hao Zhou , Zhehao Li, Lili Qiu, Swarun Kumar, Jie Xiong
UIST 25	<i>SignGlass: First-Person View Comprehensive and Generalizable ASL Translation Using Wearable Glasses.</i> Yongxiang Cai, Taiting Lu, Zhenghao Li, Hao Zhou , Kenneth DeHaan, Xuhai Xu, Mahanth Gowda, Yincheng Jin
CHI 25	<i>BallistoBud: Heart Rate Variability Monitoring Using Earbud Accelerometry for Stress Assessment.</i> Md Saiful Islam, Md Mahbubur Rahman, Mehrab Bin Morshed, David J. Lin, Yunzhi Li, Hao Zhou , Wendy Berry Mendes, and Jilong Kuang
MobiCom 24	<i>Rethinking Orientation Estimation with Smartphone-Equipped Ultra-Wideband Chips.</i> Hao Zhou , Kuang Yuan, Mahanth Gowda, Lili Qiu, Jie Xiong
IoTDI 24	<i>ASLRing: American Sign Language Recognition with Meta-Learning on Wearables.</i> Hao Zhou , Taiting Lu, Kenneth DeHaan, Mahanth Gowda
UbiComp 24	<i>SmartDampener: An Open Source Platform for Sport Analytics in Tennis.</i> Runze Liu, Taiting Lu, Shengming Yuan, Hao Zhou , Mahanth Gowda
MobiCom 23	<i>SignQuery: A Natural User Interface and Search Engine for Sign Languages with Wearable Sensors.</i> Hao Zhou , Taiting Lu, Kristina Mckinnie, Joseph Palagano, Kenneth DeHaan, Mahanth Gowda
IoTDI 23	<i>One Ring to Rule Them All: An Open Source SmartRing Platform for Finger Motion Analytics and Healthcare Applications.</i> Hao Zhou , Taiting Lu, Yilin Liu, Shijia Zhang, Runze Liu, Mahanth Gowda
TIOT 23	<i>I Am an Earphone and I Can Hear My User's Face: Facial Landmark Tracking Using Smart Earphones.</i> Shijia Zhang, Taiting Lu, Hao Zhou , Yilin Liu, Runze Liu, Mahanth Gowda

UbiComp 22 *Learning on the Rings: Self-Supervised 3D Finger Motion Tracking Using Wearable Sensors.*
Hao Zhou, Taiting Lu, Yilin Liu, Shijia Zhang, Mahanth Gowda

HONORS & AWARDS

2025	Received Penn State Alumni Association Dissertation Award
2025	Received Samsung Research America President Award
2025	Selected in ACM MobiSys Rising Star Forum
2025	Received UIST Special Recognition for Belonging and Inclusion Award
2024	Received ACM MobiCom Student Travel Grant
2023	Participated as an Entrepreneurial Lead in NSF I-Corps Program
2023	Received ACM/IEEE IoTDI Best Paper Award for Edge IoT AI
2023	Selected for Microsoft Research Asia Stars of Tomorrow
2023	Received Penn State International Student Travel Grant
2022	Received Penn State Outstanding Teaching Assistant Award
2019	Graduated with Summa Cum Laude at the University of Mississippi

WORK EXPERIENCE

2025 - 2026	Digital Health Lab, Samsung Research America
2024	Digital Health Lab, Samsung Research America
2023	Microsoft Research Asia with Prof. Jie Xiong

MEDIA COVERAGE

2025	Penn State Alumni Association Dissertation Award: Penn State News
2023	OmniRing: Hackster and DeepTech

INVITED TALKS

Inferring Human Behavior and Health from Imperfect Wearable Signals

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|-----------------|----------|
| • UW-Madison CS | Mar 2026 |
| • UT Dallas CS | Feb 2026 |

- UMBC CSEE Feb 2026
- UAB CS Feb 2026
- Penn State EE 497 Dec 2025

Pose Estimation with Wearables and Its Applications in Sign Languages

- Samsung Research America Jun 2024

Self-Supervised 3D Finger Motion Tracking using Wearable Sensors

- Microsoft Research Asia May 2023

TEACHING

Teaching Assistant @ Penn State

- CMPEN/EE 362: Communication Networks SP 2025
- CSE/EE 559: Wireless and Mobile Sensing in the age of IoT FA 2024
- CMPEN 462: Wireless Communication Systems and Security SP 2022, SP 2024

SERVICE

Program Committee

- ACM MobiCom S3 Workshop 2024
- ACM MobiCom Artifact Evaluation 2024
- ACM MobiSys Artifact Evaluation 2024

Invited Reviewer

- ACM UbiComp/IMWUT 2022 - 2026
- IEEE Transactions of Mobile Computing 2024 - 2026
- ACM CHI 2026
- ICLR 2024 - 2026
- ICML 2026
- NeurIPS 2026
- ACM Transactions on Sensor Networks 2024